North American Journal of Fisheries Management 38:549–564, 2018
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ISSN: 0275-5947 print / 1548-8675 online
DOI: 10.1002/nafm.10048

ARTICLE

Genetic Population Structure of Shoal Bass within their Native Range

Andrew T. Taylor*
Department of Natural Resource Ecology and Management, Oklahoma State University, Stillwater, Oklahoma 74078, USA

Michael D. Tringali
Florida Fish and Wildlife Conservation Commission, Fish and Wildlife Research Institute, St. Petersburg, Florida 33701, USA

Steven M. Sammons
School of Fisheries, Aquaculture, and Aquatic Sciences, Auburn University, Auburn, Alabama 36849, USA

Travis R. Ingram
Georgia Department of Natural Resources, 2024 Newton Road, Albany, Georgia 31701, USA

Patrick M. O'Rouke
Georgia Power Company, 2480 Maner Road Southeast, Atlanta, Georgia 30339, USA

Douglas L. Peterson
Warnell School of Forestry and Natural Resources, University of Georgia, Athens, Georgia 30602, USA

James M. Long
U.S. Geological Survey, Oklahoma Cooperative Fish and Wildlife Research Unit, Department of Natural Resource Ecology and Management, Oklahoma State University, Stillwater, Oklahoma 74078, USA

Abstract

Endemic to the Apalachicola River basin of the southeastern USA, the Shoal Bass *Micropterus cataractae* is a fluvial-specialist sport fish that is imperiled because of anthropogenic habitat alteration. To counter population declines, restorative stocking efforts are becoming an increasingly relevant management strategy. However, population genetic structure within the species is currently unknown, but it could influence management decisions, such as brood source location. Leveraging a collaborative effort to collect and genotype specimens with 16 microsatellite loci, our objective was to characterize hierarchical population structure and genetic differentiation of the Shoal Bass across its native range, including an examination of structuring mechanisms, such as relatedness and inbreeding levels. Specimens identified as Shoal Bass were collected from 13 distinct sites (*N* ranged from 17 to 209 per location) and were then taxonomically screened to remove nonnative congeners and hybrids (pure Shoal Bass *N* ranged from 13 to 183 per location). Our results revealed appreciable population structure, with five distinct Shoal Bass populations identifiable at the uppermost hierarchical level that generally corresponded with natural geographic features and anthropogenic barriers. Substructure was recovered within several of these populations, wherein differences appeared related to

*Corresponding author: andrew.t.taylor@okstate.edu
Received September 26, 2017; accepted January 24, 2018

spatial isolation and local population dynamics. An analysis of molecular variance revealed that 3.6% of the variation in our data set was accounted for among three larger river drainages, but substructure within each river drainage also explained an additional 8.9% of genetic variation, demonstrating that management at a scale lower than the river drainage level would likely best conserve genetic diversity. Results provide a population genetic framework that can inform future management decisions, such as brood source location, so that genetic diversity within and among populations is conserved and overall adaptability of the species is maintained.

The Shoal Bass *Micropterus cataractae* is a fluvial-specialist black bass species that is endemic to the Apalachicola–Chattahoochee–Flint River (ACF) basin of the southeastern USA (Williams and Burgess 1999; Sammons et al. 2015). In contrast to its congeners like the Largemouth Bass *M. salmoides* and Spotted Bass *M. punctulatus*, the Shoal Bass is rarely encountered in lentic habitats and does not support impoundment fisheries. Shoal Bass inhabit fluvial systems and are typically encountered near shoal habitats characterized by high flow velocities, bedrock outcrops, and boulder substrates (Williams and Burgess 1999; Johnston and Kennon 2007; Taylor and Peterson 2014). Populations inhabiting these fluvial habitats support popular sport fisheries that have gained national attention among angling groups (Taylor and Peterson 2014). However, the Shoal Bass is considered "vulnerable" by the American Fisheries Society's Endangered Species Committee, indicating that the species could become threatened with extinction throughout all or a significant portion of its range (Jelks et al. 2008). State management agencies in Alabama, Florida, and Georgia have recognized Shoal Bass as a species of conservation interest at various priority levels, but the species has not been petitioned for federal protection (Taylor and Peterson 2014; Sammons et al. 2015). A growing interest in Shoal Bass as a sport fish and as a species of concern has spurred research and management during the past decade (Birdsong et al. 2010; Tringali et al. 2015a).

Although only recognized as a species in 1999, Shoal Bass populations are presumed to have suffered dramatic losses throughout their historic native range in the ACF basin (Williams and Burgess 1999). The serial impoundment of the Chattahoochee River (Williams and Burgess 1999) has fragmented populations (Dakin et al. 2015; Sammons and Early 2015) and led to hydrologic and thermal changes downstream of dams that have negatively affected recruitment and persistence (Long and Martin 2008). The species has been largely extirpated from the Chattahoochee River south of Atlanta, Georgia, and is currently found only in a few large tributaries and in isolated locations below some main-stem dams (Sammons and Maceina 2009; Katechis 2015). In contrast, populations in the more rural and mostly undammed Flint and Chipola rivers appear relatively intact (Sammons et al. 2015). However, the invasion of previously allopatric congener species following unauthorized fish stockings now threatens Shoal Bass conservation throughout the species' range. Nonnative congeners, such as the Spotted Bass, Alabama Bass *M. henshalli*, and Smallmouth Bass *M. dolomieu*, have negatively affected Shoal Bass populations through introgressive hybridization (Alvarez et al. 2015; Dakin et al. 2015; Tringali et al. 2015a) and possible interspecific competition (Sammons 2012; Goclowski et al. 2013).

To address declining trends, resource managers have been actively managing and restoring Shoal Bass populations. Where Shoal Bass populations are robust enough to support sport fisheries, management tools such as daily creel and size limits have generally been effective at conserving stocks (Ingram and Kilpatrick 2015; Woodside et al. 2015; Sammons 2016). In other areas, stocking has been used to restore or supplement Shoal Bass populations. For example, the Georgia Department of Natural Resources (GADNR) has stocked Shoal Bass fingerlings in the lower Flint River between Lake Blackshear and Lake Worth since 1972 to supplement natural recruitment (T. Ingram, unpublished data). From 2003 to 2008, the National Park Service (NPS) and GADNR stocked Shoal Bass fingerlings in the upper Chattahoochee River below Morgan Falls Dam to re-establish a population that was depleted by the coldwater releases of an upstream dam (Long and Martin 2008; Porta and Long 2015). In both of these stocking efforts, broodfish were obtained from downstream areas within the recipient river basin. More recently, restorative stockings in the Chattahoochee River below Morgan Falls Dam (2011–present) as well as in Chattahoochee River tributaries of eastern Alabama (2009–2010) used broodstock from the lower Flint River (P. O'Rouke, unpublished data; S. Rider, Alabama Department of Conservation and Natural Resources, personal communication). Changes in brood source location to a different river drainage were made because of challenges in obtaining broodstock from the Chattahoochee River basin, and available taxon-level genetic analyses (e.g., Kassler et al. 2002) suggested little variation within the Shoal Bass. Despite the varied success of these efforts, stocking to sustain and restore Shoal Bass populations remains a relevant management tool, prompting consideration of the parental sources of stocked individuals and the potential consequences of translocating genomes.

Natural metapopulation structure of Shoal Bass is not known, but recent studies of movement patterns and demographics suggest that population-level differences occur across the species' native range. Studies of individual movements suggest that individuals in tributaries of the Chattahoochee and Flint River basins are more sedentary, exhibiting philopatry (Stormer and Maceina 2009; T. Ingram, unpublished data). Shoal Bass in free-flowing reaches of the main-stem Flint River move long distances (≤200 km; Sammons 2015) to spawn, forming large spawning aggregations at a few particular locations (Ingram et al. 2013; Taylor and Peterson 2015). In contrast, Shoal Bass situated between two main-stem dams on the Chattahoochee River were relatively sedentary (Sammons and Early 2015), suggesting an effect of anthropogenic barriers on individual movements. A marked gradient in Shoal Bass population demographics is also apparent among populations across the species' native range. Shoal Bass in the upper Chattahoochee River basin are long-lived (up to 14 years) and slow-growing, and they experience low annual mortality (20%; Porta and Long 2015; Taylor 2017). Conversely, Shoal Bass in the Flint and Chipola River basins have shorter life spans (up to 11 years), grow rapidly, and exhibit higher total annual mortality (49–57%; Ingram and Kilpatrick 2015; Woodside et al. 2015; Sammons 2016). Whether these observed differences in movement patterns and demographics represent clinal variation, local adaptation, or are the result of habitat alteration is currently unclear because these basins differ in their latitude, elevation, natural habitats, and degree of anthropogenic alteration. A better understanding of rangewide population structure and genetic variation may guide future management efforts and help to ensure that any existing adaptive and evolutionary potential is preserved.

There is growing consensus among fisheries biologists regarding the importance of conservation of genetically distinct populations to maintain genetic variation in a species (Hallerman 2003), which requires the maintenance of genetic diversity both within and among populations (Meffe 1986). Of particular concern is the loss of advantageous local adaptations and reduced capacity to evolve in response to a changing environment (Reisenbichler 1997). Maintaining genetic diversity and adaptive fitness is an especially acute issue for threatened or imperiled species that may already be losing genetic diversity because of small population sizes, inbreeding depression, and random genetic drift (Meffe 1986). However, when gene flow among genetically dissimilar populations is artificially increased through stocking, outbreeding depression and associated loss of both population and individual fitness can become unintended consequences (Lynch 1991; Hallerman 2003; Goldberg et al. 2005).

Currently, restorative stocking of Shoal Bass is relatively localized and limited in scale, but could expand in scope in the near future. With these needs in mind, our objective was to characterize hierarchical population structure and genetic differentiation within the native range of the Shoal Bass. Examining structure in a hierarchical fashion allowed determination of the spatial scale(s) at which management actions would best conserve genetic diversity. For each population identified, we summarized genetic diversity measures and examined relatedness and inbreeding levels, which provided insights into population structuring mechanisms and overall conservation genetic status. Results can inform management actions, like supplemental stocking, so that inbreeding and outbreeding depression can be avoided and natural genetic compositions within the species can be maintained.

METHODS

Study area and sampling.— To comprehensively survey genetic variation within Shoal Bass, we obtained specimens from a subset of streams that spanned the species' native range. Shoal Bass were targeted using electrofishing and hook-and-line angling, and a fin clip from each specimen was stored in an individually labeled vial of 95% nondenatured ethanol. Specimens identified as Shoal Bass in the field were considered "putative" Shoal Bass prior to taxonomic screening with genetic approaches because of ongoing introgressive hybridization with nonnative congener species in some populations (Alvarez et al. 2015; Dakin et al. 2015; Tringali et al. 2015a), and specimens that were identified as potential nonnative species or hybrids were removed prior to taxonomic screening. Specimens were collected at individual localities that spanned approximately 100 m to 5 km in stream length. Localities were grouped into a number of "sites" based on their geographic proximity to other localities and potential barriers to gene flow (e.g., dams) between localities (Figure 1). Within sites, specimens were ordered from upstream to downstream based on their sampling locality, and specimens collected within a given locality were also sorted by sampling year (oldest to newest).

Molecular analyses.— Sixteen di-nucleotide microsatellite DNA markers developed to amplify *Micropterus* (*Msaf 05*, *06*, *08*, *09*, *10*, *12*, *13*, *17*, *22*, *24*, *25*, *27*, *28*, *29*, *31*, and *32*; Seyoum et al. 2013) were used to discern species composition, explore population structure, and quantify genetic differentiation. The conformance of these markers to Hardy–Weinberg expectations has been previously established in *Micropterus* species (Seyoum et al. 2013). All assay procedures, equipment, fragment sizing, binning, and scoring protocols used in the in-house development of these markers were also used in the current work. Specifically, we isolated genomic DNA from fin

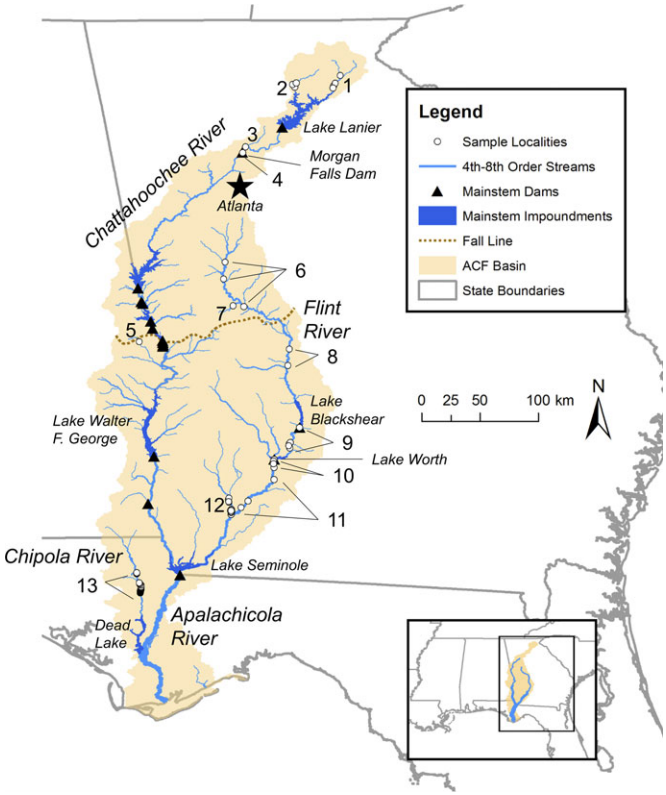

FIGURE 1. Collection localities for Shoal Bass specimens that were genotyped with 16 microsatellite DNA loci and used to characterize genetic population structure across the species' native range in the Apalachicola–Chattahoochee–Flint River (ACF) basin. Site numbers correspond to locality data in Table 1. [Color figure can be viewed at afsjournals.org.]

clips by using the Puregene DNA Purification Kit (Gentra Systems). Six multiplex PCRs were used to amplify microsatellites, with the following parameters used for all loci: 94°C for 2 min; 35 denaturation cycles of 94°C for 30 s, annealing at 58°C for 30 s, and extension at 72°C for 30 s; and a final extension at 72°C for 10 min. Using an Applied Biosystems 3130 XL Genetic Analyzer, capillary electrophoresis was performed on PCR products containing 13 μL of formamide (denaturation for 4 min at 95°C) and GeneScan ROX 500 size standard (Applied Biosystems). Negative-control PCRs were included in all plates. Laboratory protocol required any plate or individual reaction showing signs of contamination to be repeated. To minimize specimen mistypings caused by allele dropout, homozygous genotypes having peak amplitudes below a predetermined threshold (100 relative fluorescence units) were not scored. For specimens in which the majority of loci were successfully scored, a single attempt was made to resolve nonscored loci via reassay. Determination of allele length variants was performed in GeneMapper version 4.0 (Applied Biosystems). All GeneMapper standard curve and specimen fragment bin assignments were independently verified by experienced readers. Some putative Shoal Bass genotypes (~24%) contained missing data at one or more loci, but all genotypes were retained for analysis because their removal could have impeded our ability to discern population signals.

Reference specimens.— Reference genotypes representing pure specimens existed for congener taxa but not for Shoal Bass across their native range; thus, we determined suitable Shoal Bass reference genotypes from our putative specimens by using a genetic clustering analysis. Reference genotypes for the following six congener taxa relevant to the ACF basin (Taylor and Peterson 2014; Freeman et al. 2015) were obtained from established sources and provided by the Florida Fish and Wildlife Research Institute: Largemouth Bass × Florida Bass *M. s. floridanus* intergrades, Alabama Bass, Spotted Bass, "Choctaw Bass" *M.* sp. cf. *punctulatus*, Smallmouth Bass, and "Bartram's Bass" *M.* sp. cf. *cataractae* (for locality data, see Tringali et al. 2015a). All congeneric reference specimens were assayed and scored on two separate occasions by independent readers (Alvarez et al. 2015; Tringali et al. 2015a) to confirm genotype consistency. We included all reference congener genotypes and all putative Shoal Bass genotypes in a preliminary clustering analysis that determined how reference congener taxa aligned with genetic clusters and which Shoal Bass genotypes would provide suitable references for rangewide comparisons. We then used STRUCTURE (Pritchard et al. 2000) and the approaches detailed in the following sections to identify the optimal number of genetic clusters (K) within this combined data set. We performed five independent STRUCTURE runs that did not incorporate any a priori grouping information for each K ranging from 1 to 15. We then estimated the optimal K-value using the maximum value obtained from a suite of four supervised estimators (MedMeaK, MaxMeaK, MedMedK, and MaxMedK) that disregarded "spurious clusters" that failed to obtain a mean or median membership coefficient threshold of at least 0.50 within at least one grouping of genotypes (Puechmaille 2016). From the optimal K clustering assignment, we selected a number of Shoal Bass reference genotypes representative of the entire sampled range that had individual genomic proportion (q-value) assignments of at least 0.980 to Shoal Bass-affiliated clusters. Studies that employed similar markers and methodologies to assess hybridization reported thresholds for pure Shoal Bass at q of 0.980 or greater in the Chipola River (Tringali et al. 2015a) and q greater than 0.982 in the lower Flint River (Alvarez et al. 2015).

Taxonomic screening.— The reference genotypes for Shoal Bass and the six congener taxa were used to perform a formal taxonomic assignment to identify "pure" Shoal Bass genotypes, wherein each putative Shoal Bass genotype was proportionally assigned (i.e., q-values) to each taxon based on the allelic frequencies of reference

genotypes. To complete the taxonomic assignment, we employed a Bayesian clustering approach in STRUCTURE version 2.3.4 (Pritchard et al. 2000) and accessory programs. STRUCTURE proportionally assigned individual genotypes to a given number of genetic clusters (i.e., K) based on linkage equilibrium and conformance to Hardy–Weinberg equilibrium (Pritchard et al. 2000).

Using STRUCTURE, we assumed the admixture ancestry model and independent allele frequencies, and we used a burn-in of 20,000 and 200,000 Markov chain–Monte Carlo (MCMC) repetitions for each run. The "PopFlag" option was employed so that genomic proportions for putative Shoal Bass genotypes were estimated using the allele frequencies from the reference genotypes of each taxon. We conducted 20 independent runs at the optimal K-value identified in the previous analysis used to identify Shoal Bass reference specimens. We then input those 20 runs into STRUCTURE HARVESTER web version 0.6.94 (Earl and vonHoldt 2012) to obtain input files for CLUMPP version 1.1.2, which provided optimal alignment of independent STRUCTURE runs with cluster matching and permutation (Jakobsson and Rosenberg 2007). Final q-value assignment in CLUMPP was obtained using the G' pairwise matrix similarity statistic and the "LargeK-Greedy" algorithm for 1,000 randomly sequenced runs. To determine the q-value threshold for "pure" Shoal Bass individuals, we followed Littrell et al. (2007).

Population structure.— Species distributed across geographic space are often subdivided into populations, with genetic differentiation occurring both among and within populations (Excoffier et al. 1992). To characterize differences among and within populations, we assessed hierarchical genetic structure of "pure" Shoal Bass by using a similar Bayesian clustering approach implemented in STRUCTURE (Pritchard et al. 2000). Briefly, we first estimated the optimal K-value at the uppermost hierarchical level of structure, delineating large-scale, population-level differences in our data set. We then examined whether substructure existed within each of the populations identified at the uppermost level by estimating K within each of those populations. For all STRUCTURE runs, we assumed the admixture ancestry model and correlated allele frequencies, and we used a burn-in of 20,000 and 200,000 MCMC repetitions for each run. We estimated K within each hierarchical level of genetic structure using the maximum value of four supervised estimators (MedMeaK, MaxMeaK, MedMedK, and MaxMedK) that disregarded any "spurious clusters" failing to obtain a mean or median membership coefficient threshold of at least 0.50 for at least one site (Puechmaille 2016).

We proceeded in a hierarchical fashion, first estimating the optimal K-value at the uppermost hierarchical level. To reduce the effect of uneven sampling across all sites, we used supervised estimators that accurately identified hierarchical structure in unevenly sampled data sets (Puechmaille 2016). We also analyzed a geographically subsampled data set that contained approximately equal numbers of randomly selected genotypes at each site (maximum of 20). For each data set, we ran 10 independent runs of $K = 1$–15 without any a priori sample site information. The data set producing the largest K-value was then adopted to explore the secondary level of hierarchical genetic structure among sites that had a mean membership coefficient of at least 0.50 to a given cluster. Sites with mean membership coefficients less than 0.50 were considered areas of mixture at the uppermost level and were not explored further. To explore the secondary level of hierarchical genetic structure, we conducted 10 independent runs of $K = 1$–5 for each combination of sites dominated by a given genetic cluster. If no structure was detected, we also performed the analysis using sample sites as prior information ("LocPrior" option), which allowed STRUCTURE models to more easily detect existing structure among sites even if population signals were weak (Pritchard et al. 2000). Final q-value assignments were obtained in CLUMPP with the same settings described previously. For each hierarchical level's optimal K, we recorded the mean and SD of the log probability, $L(K)$, which provided a simple measure of model fit.

Structuring mechanisms.— Because STRUCTURE models assume linkage equilibrium and Hardy–Weinberg equilibrium within populations (Pritchard et al. 2000), differences in homozygosity, allelic diversity, private alleles, and mating patterns (i.e., relatedness and inbreeding levels) can provide insights into structuring mechanisms. For each cluster or "stock" identified, we determined the number of private alleles and the mean number of alleles (A), effective number of alleles (A_e), expected heterozygosity (H_e), and observed heterozygosity (H_o) by using GenAlEx version 6.502 (Peakall and Smouse 2006). We also calculated the mean pairwise relatedness (r) and mean inbreeding coefficient (F) within each cluster. These metrics were calculated in COANCESTRY (Wang 2011) using 1,000 bootstraps and 100 control samples to estimate r with the TrioML estimator (Wang 2007) and to estimate F with the Ritland estimator (Ritland 1996). We also tested for differences in mean r and F between subpopulations contained within a higher-level population by using 1,000 bootstraps to generate 95% confidence intervals used to determine significance at $P \leq 0.05$ (Wang 2011). We visualized proportional structure of r in all identified populations using the following general kinship categories: 0.0% = unrelated; 0.1–6.2% = second (or greater) cousins; 6.3–12.5% = first cousins or half-niece/nephew; 12.6–25.0% = grandchild or niece/nephew; 25.1–50.0% = approaching child or sibling; and 50.1–100.0% = inbred.

Genetic differentiation.— We performed an analysis of molecular variance (AMOVA; Excoffier et al. 1992) in

ARLEQUIN version 3.5.1.3 (Excoffier and Lischer 2010) to estimate the amount of genetic variation present at the following hierarchical levels (largest to smallest): among major river drainages (Chattahoochee, Flint, and Chipola rivers), among sites within major river drainages, and within sites. We used the locus-by-locus AMOVA methodology to produce a global AMOVA from a weighted average over all loci. We also computed pairwise fixation index values between sites (i.e., F_{ST}; Wright 1951; Weir and Cockerham 1984), which provided a commonly used measure of genetic differentiation. Negative F_{ST} values were interpreted as no differentiation ($F_{ST} = 0$). Tests of significance for the global AMOVA and pairwise F_{ST} values were calculated based on 10,000 permutations at $P \leq 0.05$.

RESULTS

From 2005 to 2015, our collaborative efforts yielded 829 putative Shoal Bass genotypes from 13 sites spanning the native range of the species (Table 1; Figure 1). Genotypes from several sites could have been influenced by local stocking efforts, as indicated in Table 1. The lower Flint River below Lake Blackshear (site 9) and the Chipola River (site 13) were each represented by at least 150 specimens, whereas the remaining sites were represented by 17–77 specimens each.

Reference Specimens

The six congener taxa included in our analyses were represented by approximately 40–60 reference genotypes each, although Bartram's Bass was represented by only 15 samples (Table 2). Among the six congener reference taxa and all putative Shoal Bass genotypes, the optimal number of clusters was $K = 10$, wherein the six congener taxa were each represented by distinct genetic clusters and putative Shoal Bass genotypes contained four additional, unique genetic clusters. The four Shoal Bass-affiliated clusters identified were likely an artifact of the large sample size of putative Shoal Bass genotypes rather than reflecting taxon-level differences (sensu Puechmaille 2016); thus, we grouped the four Shoal Bass clusters into one, from which the $q \geq 0.980$ selection criterion was applied to attain 55 Shoal Bass genotypes suitable to serve as references (mean $q = 0.992$; range = 0.980–0.994).

Taxonomic Screening

The formal taxonomic assignment using the six congener reference taxa plus the Shoal Bass references was conducted at $K = 7$ (one Shoal Bass cluster plus six other distinct clusters that aligned with congener taxa). The resulting classification threshold for pure Shoal Bass within our putative genotypes was $q \geq 0.988$. Based on this threshold, 644 genotypes (individual sites containing a range of 13–183 genotypes) were identified as "pure" Shoal Bass suitable for hierarchical analyses of population structure and genetic variation (Figure 2).

Population Structure

At the uppermost hierarchical level, the complete data set of pure Shoal Bass genotypes ($N = 644$) was estimated to contain $K = 5$ clusters, whereas the geographically subsampled data set of evenly represented genotypes was estimated to contain $K = 4$ clusters. The complete data set at $K = 5$ provided an assignment with clusters representing the Chattahoochee and Chestatee rivers above Lake Lanier (sites 1 and 2); Little Uchee Creek (site 5); the upper Flint River and Big Lazer Creek (sites 6–8); the lower Flint River (sites 9 and 10); and the Chipola River (site 13; Figure 3A). Sites 3, 4, 11, and 12 were areas of mixture, with no clear assignment to any cluster (mean membership coefficients <0.50). The subsampled data set at $K = 4$ produced a similar grouping, with distinct clusters in Little Uchee Creek (site 5), another cluster containing the upper Flint River and Big Lazer Creek (sites 6–8), and all other sites assigned to a mixture of two other clusters. Because the complete data set at $K = 5$ provided an improved ability to discern population signals among sites, it was adopted to explore the secondary level of hierarchical genetic structure. Genomic proportions at $K = 5$ illustrated that some populations had noticeably low connectivity to neighboring populations (Little Uchee Creek, Chipola River, and upper Flint River; Figure 4). In contrast, assignments from the lower Flint River population indicated high connectivity to the upper Chattahoochee River and upper Flint River populations.

At the secondary hierarchical level, substructure was evident in three groups (Figure 3B). Substructure was detected ($K = 2$) within the Chipola River (site 13) without prior information; however, the substructure recovered did not correspond with spatial position within the river but rather corresponded to sampling year (2007–2009 versus 2010). Substructure ($K = 2$) was also recovered between the upper Chattahoochee River (site 1) and Chestatee River (site 2) as well as between Big Lazer Creek (site 7) and main-stem upper Flint River sites (sites 6 and 8); however, substructure was only evident in these regions when site information was used as a model prior. No additional structure ($K = 1$) was recovered in Little Uchee Creek (site 5) or in the lower Flint River sites (sites 9 and 10), regardless of whether site information was used as prior information. We did not investigate genetic structure beyond the second hierarchical level, as population signals were much weaker at the secondary level and, in many instances, the subpopulation boundaries corresponded with sampling sites (i.e., smallest spatial grain). Mean estimates of $L(K)$ indicated best fit at the uppermost

TABLE 1. Specimens collected for analysis of Shoal Bass genetic structure across their native range, including site number (see Figure 1), location description, major river drainage (CHA = Chattahoochee River; FLI = Flint River; CHI = Chipola River), collection years, number (*N*) of putative Shoal Bass collected, and number (*N*) of genetically "pure" Shoal Bass identified after taxonomic screening.

Site	Location	Drainage	Collection years	Putative *N*	Pure *N*
1	Upper Chattahoochee River	CHA	2005, 2013	69	59
2	Chestatee River	CHA	2013	42	32
3	Big Creek	CHA	2005, 2015	77	23
4	Chattahoochee River below Morgan Falls Dam[a]	CHA	2005, 2015	42	19
5	Little Uchee Creek[b]	CHA	2008	22	22
6	Upper Flint River above Fall Line	FLI	2009, 2011	32	24
7	Big Lazer Creek	FLI	2013	18	16
8	Upper Flint River below Fall Line	FLI	2011	17	13
9	Lower Flint River below Lake Blackshear[a]	FLI	2009–2011	159	137
10	Lower Flint River below Lake Worth[a]	FLI	2009–2011	28	24
11	Lower Flint River	FLI	2009, 2011	50	35
12	Ichawaynochaway Creek	FLI	2010–2011	64	57
13	Chipola River	CHI	2007–2010	209	183
Total				829	644

[a]Stocking activities in the area were prior to, or contemporary with, specimen collection.
[b]Stocking activities in the area took place after specimen collection.

TABLE 2. Common name, scientific name, and number of specimens included as reference genotypes for the taxonomic screening of microsatellite DNA genotypes used in analysis of Shoal Bass genetic structure across the species' native range.

Common name	Scientific name	*N*
Largemouth Bass × Florida Bass	*Micropterus salmoides* × *M. s. floridanus*	62
Alabama Bass	*M. henshalli*	63
Spotted Bass	*M. punctulatus*	69
"Choctaw Bass"[a]	*M.* sp. cf. *punctulatus*	56
Smallmouth Bass	*M. dolomieu*	41
"Bartram's Bass"[b]	*M.* sp. cf. *cataractae*	15
Shoal Bass	*M. cataractae*	55
Total		361

[a]"Choctaw Bass" is a provisional form that has been recommended for species recognition (Tringali et al. 2015a).
[b]"Bartram's Bass" is a provisional form that has been recommended for species recognition (Freeman et al. 2015).

hierarchical level ($K = 5$) and less fit in groupings among the secondary hierarchical level (Table 3).

Structuring Mechanisms

No large deviations from Hardy–Weinberg equilibrium occurred, as cluster-specific H_o did not deviate by more than |0.031| from H_e (Table 4). At the uppermost hierarchical level, allelic diversity was relatively low in the Little Uchee Creek ($A_e = 1.378$) and Chipola River ($A_e = 1.958$) clusters. At the secondary level, the Chestatee River cluster also had relatively low allelic diversity ($A_e = 1.995$). Heterozygosity was markedly lower ($H_o = 0.199$) in the Little Uchee Creek cluster. In contrast, the upper Flint River cluster had some of the highest measures of allelic diversity ($A_e = 2.396$) and heterozygosity ($H_o = 0.370$). The Chipola River population had the highest number of private alleles, with 4 in the 2007–2009 sample years and an additional 4 in the 2010 sample year. Mean pairwise relatedness was highest in Little Uchee Creek ($r = 0.162$), followed by the 2010 subgrouping in the Chipola River ($r = 0.138$), which also had the highest mean inbreeding coefficient ($F = 0.131$; Table 5). Tests for significant differences in mean r and F between subpopulations nested within an upper-level population revealed one significant difference—pairwise relatedness between the upper Chattahoochee River and the Chestatee River ($r = 0.082$ versus 0.107, respectively). Proportional structure of pairwise relatedness was generally similar across all sites, with a few notable exceptions: Little Uchee Creek had a relatively high percentage of pairwise relationships indicative of inbreeding (18%, compared to approximately 1% elsewhere), Big Lazer Creek had a relatively high percentage of pairwise relationships approximating first cousins (23%, compared to 10–15% elsewhere), and the 2010 grouping in the Chipola River had a relatively low percentage of pairwise relationships that were unrelated (31%, compared to approximately 50% elsewhere; Figure 5).

Genetic Differentiation

The global AMOVA indicated that a significant majority (87.4%; $P < 0.0001$) of genetic variation was located within sites (Table 6). Significant variation was also attributable to differences among sites within river drainages

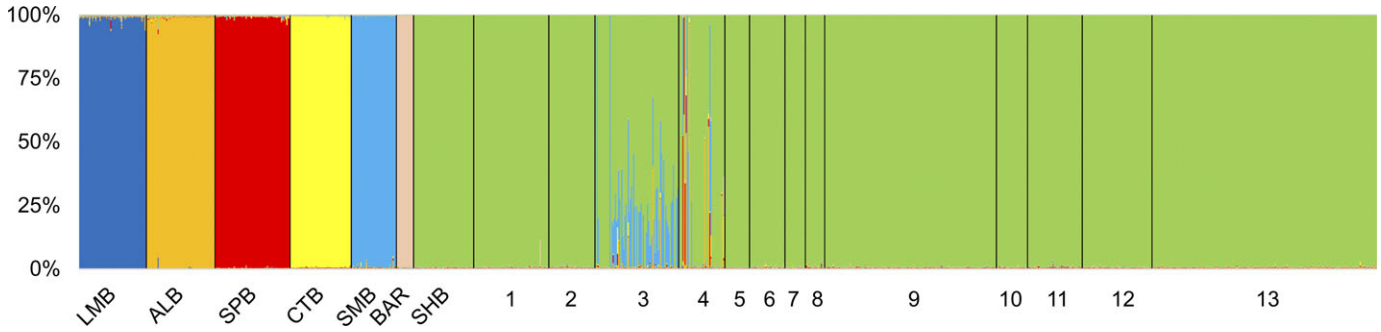

FIGURE 2. Taxonomic assignment of 829 putative Shoal Bass that were collected from 13 sites (see Table 1) and genotyped with 16 microsatellite DNA loci. Assignments were made using the allele frequencies of reference genotypes for seven *Micropterus* taxa (LMB = Largemouth Bass × Florida Bass intergrades; ALB = Alabama Bass; SPB = Spotted Bass; CTB = Choctaw Bass; SMB = Smallmouth Bass; BAR = Bartram's Bass; SHB = Shoal Bass). Distinct genetic clusters are represented by different shading (print version) or different colors (online version), and each individual's proportional assignment to those clusters is represented within a single vertical bar. [Color figure can be viewed at afsjournals.org.]

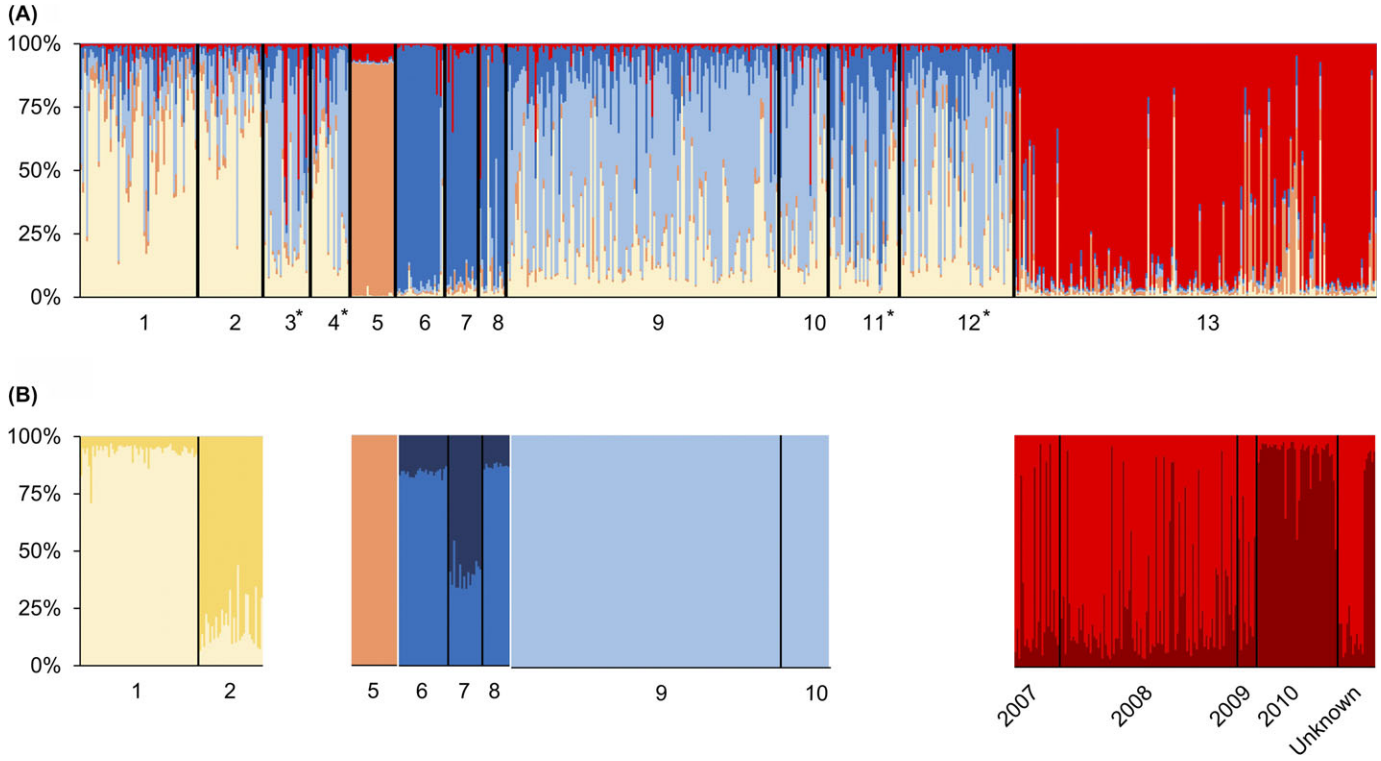

FIGURE 3. Hierarchical population structure within 644 pure Shoal Bass that were collected from 13 sites across the species' range (see Table 1) and genotyped with 16 microsatellite DNA loci. **(A)** The uppermost level of hierarchical structure contained five genetic clusters. **(B)** Structure at the secondary level was also apparent in some populations. Sites at the uppermost level that did not have a mean membership coefficient of at least 0.50 to a given cluster (designated by asterisks) were considered mixtures of multiple clusters and were not investigated for lower-level structure. Distinct genetic clusters are represented by different shading (print version) or different colors (online version), and each individual's proportional assignment to those clusters is represented within a single vertical bar. [Color figure can be viewed at afsjournals.org.]

(8.9%; $P < 0.0001$), and a small but significant remainder of the variation was attributable to differences among river drainages (3.6%; $P = 0.021$). Pairwise F_{ST} values indicated that the following sites were significantly ($P \leq 0.05$) differentiated from all other sites: upper Chattahoochee River (site 1), Chestatee River (site 2), Big Creek (site 3), Little Uchee Creek (site 5), Big Lazer Creek (site 7), and Chipola River (site 13; Table 7). The Little Uchee Creek (site 5) population consistently had the highest pairwise F_{ST} values, ranging from 0.20 to 0.36. The Chipola River (site 13) population also had high pairwise F_{ST} values ranging from 0.07 to 0.19.

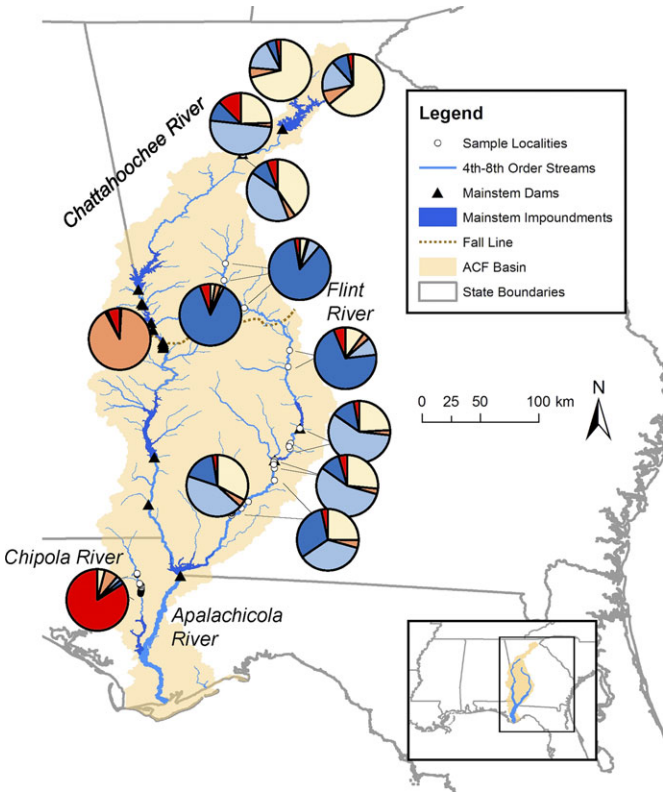

FIGURE 4. Overall genomic proportions of Shoal Bass from each of 13 sites across the species' range in the Apalachicola–Chattahoochee–Flint River (ACF) basin. Assignments correspond to five distinct population clusters obtained at the uppermost level of hierarchical genetic structure. [Color figure can be viewed at afsjournals.org.]

DISCUSSION

Recent efforts to manage and conserve black basses have increasingly focused on understanding genetic diversity within species (Shaw 2015). Our results demonstrate that considerable—and previously undocumented—genetic structure exists within Shoal Bass across their native range. Five populations were delineated at the uppermost hierarchical level of genetic structure, with substructure apparent within three of these populations. Results not only provide insight into the ecology of the species and the conservation genetic status of populations but also identify population boundaries that can be used to inform future management and restoration efforts.

The population structure at the uppermost hierarchical level best aligned with natural landscape features that restrict connectivity, although anthropogenic alterations may have amplified differentiation in some cases. The Chipola River population had high levels of genetic differentiation when compared to populations in the Chattahoochee and Flint rivers, highlighting its relative distinctiveness and conservation value. Dead Lake, a swampy natural lake situated between shoal habitats in the Chipola River and the larger Apalachicola River (Tanner 1966), appears to isolate the Chipola River population. Although a few Shoal Bass have been reported nearby in the Apalachicola River below Lake Seminole, modern connectivity with the Chipola River population is unlikely because Shoal Bass have been functionally eliminated from the Apalachicola River by dredging of shoals and other habitat alterations (Williams and Burgess 1999). Little Uchee Creek is one of the most isolated populations of Shoal Bass known, located in a 600-m reach of shoal habitat bounded by two 4-m-tall waterfalls (Stormer and Maceina 2009). Downstream, virtually no shoal habitat exists, as Uchee Creek becomes a typical coastal plain stream with sandy substrate and shallow depths for 42 km downstream to its confluence with the Chattahoochee River, which now lies within Walter F. George Reservoir (Katechis 2015). In the Flint River basin, a natural high-gradient transition zone between the Piedmont and Southeastern Plains ecoregions, known as the Fall Line, could represent a natural barrier for upstream gene flow. The upper Flint River population above the Fall Line (fixed for a unique allele at one locus) was estimated to contain few lower Flint River population alleles, but below the Fall Line alleles from the lower Flint River population increased in prevalence. However, anthropogenic activities like the impoundment of Lake Blackshear in 1930, along with the ongoing supplemental stocking activities in the Flint River below Lake Blackshear, could have increased the differentiation between the two populations recovered in

TABLE 3. Bayesian clustering model specifications and fit statistics used to investigate hierarchical Shoal Bass genetic structure across the species' native range, including hierarchical level of genetic structure (level), genetic cluster (site numbers from Table 1), estimated number of clusters (K), whether or not site locations were used as model priors (LocPrior), mean estimated log probability of the data ($L[K]$), and SD of each mean estimated $L(K)$.

Level	Genetic cluster (site numbers)	K	Runs	LocPrior	Mean estimated $L(K)$	SD
1	All	5	10	No	−13,617.26	911.89
2	1 and 2	2	10	Yes	−1,853.21	4.05
2	5	1	10	No	−211.03	0.29
2	6, 7, and 8	2	10	Yes	−1,286.00	26.89
2	9 and 10	1	10	No	−3,237.38	0.24
2	13 (by year)	2	10	No	−3,238.13	39.65

TABLE 4. Measures of genetic diversity within Shoal Bass populations identified based on analysis of 16 microsatellite DNA loci. Levels (in brackets) indicate the hierarchy of genetic population structure, where level 1 is the uppermost hierarchical level and level 2 is secondary structure nested within level 1. Site numbers corresponding to locations on the study map (Figure 1) are given in parentheses after the site names. Measures of genetic diversity include the mean across all loci for expected heterozygosity (H_e), observed heterozygosity (H_o), number of alleles (A), and effective number of alleles (A_e). We also report the number of private alleles ($A_{private}$) and the sample size of genotypes (N).

[Level] cluster name (site number)	H_e	H_o	A	A_e	$A_{private}$	N
[1] Above Lake Lanier (1 and 2)	0.324	0.320	3.625	2.223	3	91
[2] Upper Chattahoochee River (1)	0.316	0.319	3.500	2.206	2	59
[2] Chestatee River (2)	0.324	0.324	2.875	1.995	1	32
[1] Little Uchee Creek (5)	0.189	0.199	1.625	1.378	1	22
[1] Upper Flint River (6, 7, and 8)	0.364	0.334	4.313	2.419	3	53
[2] Upper Flint River main stem (6 and 8)	0.370	0.345	3.938	2.396	2	37
[2] Big Lazer Creek (7)	0.332	0.309	3.063	2.165	1	16
[1] Lower Flint River (9 and 10)	0.309	0.313	4.375	2.308	3	161
[1] Chipola River (13)	0.327	0.310	3.313	1.958	8	183
[2] Chipola River, years 2007–2009	0.322	0.309	2.875	1.844	4	123
[2] Chipola River, year 2010	0.326	0.319	2.813	1.861	4	41
[2] Chipola River, year unknown	0.311	0.292	2.188	1.804	0	19
[1] Mixed (3, 4, 11, and 12)	0.324	0.323	4.500	2.345	1	134

the Flint River basin. A similar scenario occurred in the upper Chattahoochee River basin, wherein natural influences of increased latitude and elevation on genetic structure were confounded by the location of Lake Lanier relative to our sampling sites (see also Dakin et al. 2015). Expanded spatial coverage of specimen collections, alternative markers (e.g., mitochondrial DNA genes and single-nucleotide polymorphisms), and landscape-genetic approaches could help to disentangle the relative influences of natural and anthropogenic barriers on Shoal Bass population differentiation. Such work is planned in the future toward best informing management to mitigate the effects of anthropogenic fragmentation while conserving overall genetic diversity.

Examination of population substructure and structuring mechanisms revealed the influences of landscape-level isolation and internal population dynamics on genetic

TABLE 5. Mean pairwise relatedness (r) and mean inbreeding coefficient (F) measures within Shoal Bass populations identified based on analysis of 16 microsatellite DNA loci. Levels (in brackets) indicate the hierarchy of genetic population structure, where level 1 is the uppermost hierarchical level and level 2 is secondary structure nested within level 1. Site numbers corresponding to locations on the study map (Figure 1) are given in parentheses after site names. Sample sizes (N) differ because r was calculated with pairwise groupings of individuals, whereas F was calculated using the number of individuals. We also tested for significant differences ($P \leq 0.05$) in mean r and F between subpopulations (level 2) contained within a higher-level population (level 1). Significant comparisons are indicated by bold italics.

	r			F		
[Level] cluster name (site number)	Estimate	SD	N	Estimate	SD	N
[1] Above Lake Lanier (1 and 2)	0.078	0.129	4,095	0.000	0.141	91
[2] Upper Chattahoochee River (1)	***0.082***	0.127	1,711	−0.007	0.151	59
[2] Chestatee River (2)	***0.107***	0.163	496	0.013	0.118	32
[1] Little Uchee Creek (5)	0.162	0.220	231	−0.068	0.266	22
[1] Upper Flint River (6, 7, and 8)	0.060	0.103	1,378	0.030	0.109	53
[2] Upper Flint River main stem (6 and 8)	0.071	0.111	666	0.027	0.113	37
[2] Big Lazer Creek (7)	0.087	0.120	120	0.038	0.100	16
[1] Lower Flint River (9 and 10)	0.078	0.130	12,880	−0.010	0.101	161
[1] Chipola River (13)	0.115	0.164	13,366	0.078	0.459	164
[2] Chipola River, years 2007–2009	0.137	0.179	7,503	0.061	0.361	123
[2] Chipola River, year 2010	0.138	0.170	820	0.131	0.668	41

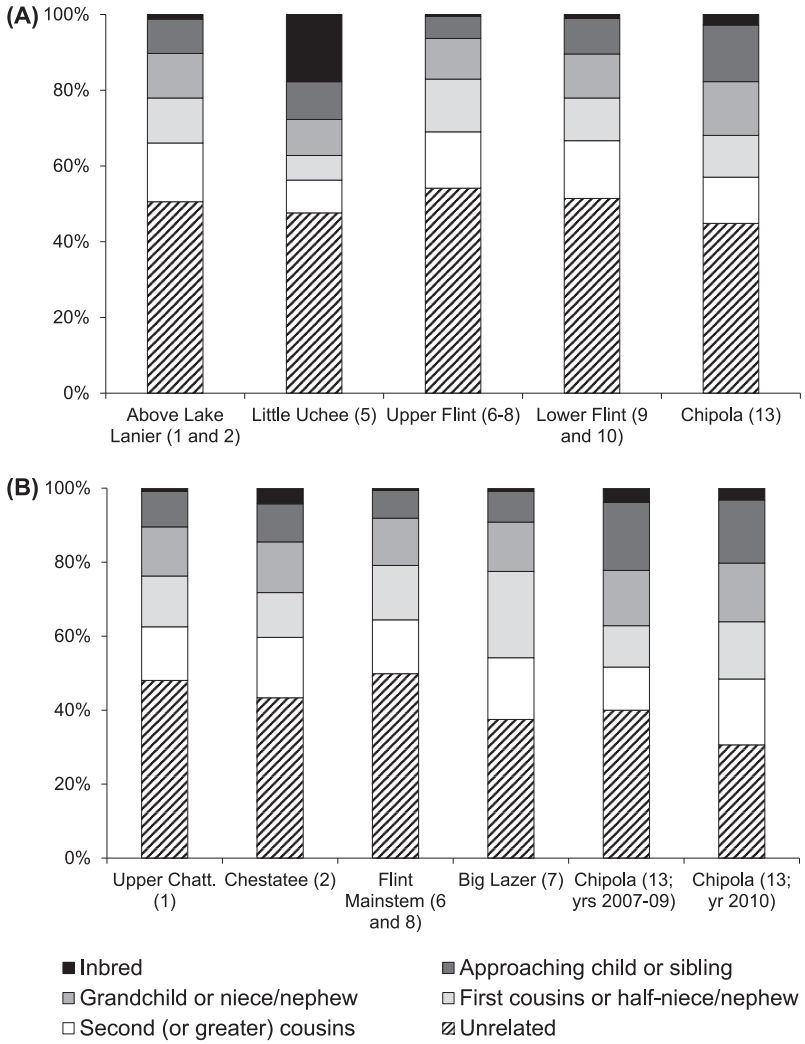

FIGURE 5. Proportional structure of pairwise relatedness (r) in identified genetic clusters (site numbers in parentheses correspond to those in Table 1; Chatt. = Chattahoochee River) of Shoal Bass across their native range at **(A)** the uppermost hierarchical level of genetic structure and **(B)** the secondary level of genetic structure. Ranges in r-values correspond to the following general kinship categories: 50.1–100.0% = inbred; 25.1–50.0% = approaching child or sibling; 12.6–25.0% = grandchild or niece/nephew; 6.3–12.5% = first cousins or half-niece/nephew; 0.1–6.2% = second (or greater) cousins; and 0.0% = unrelated.

diversity and population stability. In the Chipola River population, we discovered temporal substructure that is consistent with known recruitment variation. The 2010 Chipola River sample displayed increased relatedness measures and approximately double the amount of inbreeding from previous years' samples. Poor recruitment and year-class failures have been documented in the Chipola River population before, often in association with increased spring and summer discharge (Woodside et al. 2015). The 2010 sample was dominated by individuals belonging to the 2008 year-class, one of the first year-classes following a prolonged span of unfavorable discharge conditions from 2000 to 2006 (Woodside et al. 2015). The effect of recruitment variability in isolated populations on inbreeding levels and susceptibility to extirpation is worthy of further investigation. In the Little Uchee Creek population, lack of genetic connectivity, along with the effects of inbreeding and random genetic drift in a small population, contributed to the high levels of differentiation and the low values of H_o and A_e. Little Uchee Creek harbored the smallest known Shoal Bass population, varying from 13 to 122 estimated adults (Stormer and Maceina 2008; S. Sammons, unpublished data) and appears to now be functionally extirpated (Katechis 2015). Inbreeding can lead to reduced fitness through increased homozygosity or expression of deleterious recessive alleles, and as effective population size decreases, the effects of random genetic drift increasingly alter allele frequencies (Hallerman 2003). These processes, combined with the isolation of the Little Uchee Creek population, could explain the functional

extirpation of the population (see Gilpin and Soulé 1986). Several other isolated Chattahoochee River tributaries harbor populations that may be on a similar trajectory as Little Uchee Creek (e.g., Big Creek and several Chattahoochee River tributaries in Alabama; Stormer and Maceina 2008; Dakin et al. 2015; Taylor 2017), which is cause for conservation concern.

The results of our study have obvious implications for management and conservation of Shoal Bass populations throughout their range. Population genetic diversity serves as a "diversified portfolio" to maintain overall stability and adaptability of a species (Hallerman 2003; Schindler et al. 2010), which is particularly pertinent for species of conservation concern like the Shoal Bass (Meffe 1986). The large amount of within-site genetic variation obtained in our study mirrors that observed in populations of similar species, such as riverine and lacustrine populations of Smallmouth Bass (Borden and Stepien 2006; Hallerman et al. 2015), but management at a "site" spatial scale (i.e., stream segments) may be impractical. Conversely, only a small (3.6%) yet significant amount of genetic variation was explained by differences among three main river drainages (Chattahoochee, Flint, and Chipola rivers). Substructure within each river drainage also explained a substantial amount of genetic variation (8.9%), demonstrating that management at a scale lower than the river drainage level would likely best conserve genetic diversity across the native range of the Shoal Bass. We identified five distinct genetic populations of Shoal Bass at the uppermost hierarchical level and some appreciable substructure below that level, which could be used to define relevant management units that recognize diversity within and among populations (Barthel et al. 2010; Hallerman et al. 2015). A precautionary approach toward avoiding outbreeding depression would be to avoid stocking practices that artificially mix populations identified at the uppermost hierarchical level of structure. However, population boundaries may be justifiably disrupted to prevent diversity losses from inbreeding and random genetic drift in smaller, isolated populations like that in Little Uchee Creek.

Tributary populations, especially those fragmented from main-stem populations by dams or impounded habitats, represent some of the best opportunities for population restoration efforts. Expansive reaches of suitable habitat once occurred throughout the ACF basin prior to fragmentation by dams, which resulted in smaller, more disjunct patches of suitable habitat (Taylor et al. 2018). Thus, historical populations likely had larger census sizes and more demographic connectivity than they do at present, which could generate time lags in the response of our genetic approaches to fragmentation (see Epps and Keyghobadi 2015). If time lags are affecting our results, modern genetic connectivity among our study sites may be quite lower than our results depict (Epps and Keyghobadi 2015). For example, individual proportional assignments in the upper Chattahoochee River and lower Flint River basins illustrate moderate levels of connectivity among these populations, yet approximately a dozen main-stem dams are currently situated between these sites. Thus, the population structure uncovered in our study may not yet fully characterize the effects of large, main-stem impoundment construction in the ACF basin (which was prominent in the 1950s and 1960s), and smaller populations inhabiting tributaries that are fragmented from larger populations may be on a similar trajectory toward random genetic drift, inbreeding depression, and functional extirpation as was observed in Little Uchee Creek. For example, Big Creek harbors unique native genetic diversity but is a small, demographically disconnected population that is vulnerable to inbreeding depression (Dakin et al. 2015; Taylor 2017). Supplemental stocking may bolster population viability in fragmented tributaries, but such actions may also result in loss of unique genetic diversity if mixed with dissimilar populations.

Because stocking efforts are ongoing and are likely to increase, our results could help to refine stocking strategies that minimize the risks associated with inbreeding depression while avoiding the potential risks associated with outbreeding depression. Some Shoal Bass populations are fragmented and vulnerable to the effects of low genetic diversity and inbreeding depression (e.g., Big Creek and several other Chattahoochee River tributaries; Stormer and Maceina 2008; Dakin et al. 2015; Taylor 2017). Introgressive hybridization with nonnative congeners also appears to be a widespread threat to the genetic integrity of the Shoal Bass throughout its range (Alvarez et al. 2015; Dakin et al. 2015; Tringali et al. 2015a). Stocking efforts could serve to supplement dwindling populations, restore extirpated areas, and swamp gene pools with native alleles to combat introgression of nonnative alleles.

TABLE 6. Hierarchical analysis of molecular variance results for Shoal Bass genotyped with 16 microsatellite DNA loci. Thirteen sample sites spanned the native range of the species and were grouped into three major river drainages: the Chattahoochee, Flint, and Chipola River drainages (Table 1). Variation (%) describes the amount of genetic variation contained within each hierarchical level, and *P*-values ≤ 0.05 were considered significant.

Source of variation	Sum of squares	Variation (%)	*P*-value
Among major river drainages	191.53	3.6	0.02
Among sites within major river drainages	200.56	8.9	<0.01
Within sites	3,177.47	87.4	<0.01
Total	3,569.56	100.0	

TABLE 7. Pairwise fixation index (F_{ST}) values (Wright 1951; calculated by the method of Weir and Cockerham 1984) quantifying genetic differentiation between Shoal Bass that were sampled at each of 13 sites (see Table 1 for site names) spanning the species' native range and that were genotyped with 16 microsatellite DNA loci. Values in bold italics were considered significant at $P \leq 0.05$. Negative values are artifacts of calculation methods and were interpreted as $F_{ST} = 0$.

Site	1	2	3	4	5	6	7	8	9	10	11	12	13
1													
2	***0.03384***												
3	***0.02320***	***0.06987***											
4	***0.02192***	***0.01638***	***0.03870***										
5	***0.30018***	***0.31559***	***0.35637***	***0.29983***									
6	***0.03198***	***0.05069***	***0.03631***	0.01246	***0.30621***								
7	***0.04290***	***0.03967***	***0.06998***	***0.02994***	***0.28072***	***0.04112***							
8	***0.03024***	***0.02511***	***0.04092***	−0.00627	***0.28492***	0.00278	***0.02918***						
9	***0.04094***	***0.05044***	***0.03205***	0.00798	***0.26841***	***0.02312***	***0.05114***	0.00447					
10	***0.03925***	***0.06033***	***0.02136***	0.01147	***0.31081***	***0.02619***	***0.06781***	0.01653	0.00359				
11	***0.05256***	***0.06244***	***0.06678***	0.01213	***0.27974***	***0.01536***	***0.04756***	0.01207	***0.01778***	***0.01960***			
12	***0.03618***	***0.04757***	***0.04677***	0.00143	***0.28500***	***0.01513***	***0.04821***	0.00488	***0.01072***	0.00814	0.00672		
13	***0.12232***	***0.12416***	***0.13356***	***0.10779***	***0.19874***	***0.12395***	***0.07330***	***0.08866***	***0.13851***	***0.14254***	***0.12035***	***0.13370***	

For example, supplemental stocking to restore native Guadalupe Bass *M. treculii* in Texas streams reduced introgression with nonnative Smallmouth Bass without severely depressing native genetic diversity (Bean et al. 2013; Fleming et al. 2015; but see Littrell et al. 2007). Similar efforts to restore or supplement Shoal Bass populations are warranted and hold promise as a management and conservation tool, but careful selection of an appropriate broodstock source would be critical in determining the ultimate effectiveness of such conservation efforts. Despite the potential utility of stocking, stocking-related activities that disregard existing population boundaries could result in a loss of genetic diversity and outbreeding depression (Rhymer and Simberloff 1996; Hallerman 2003). For example, Largemouth Bass crossed from two distinct populations exhibited increased susceptibility to disease when compared to either of the nonoutcrossed native populations (Goldberg et al. 2005). Previous supplemental stocking efforts for Shoal Bass have used local broodfish, but more recent efforts have used broodfish obtained from different river basins—a practice that is less likely to maintain overall genetic diversity and more likely to result in outbreeding depression. Although no investigations of outbreeding depression have been conducted in crossed Shoal Bass stocks, F_{ST} values suggest that a crossed stock now exists below Morgan Falls Dam between the native upper Chattahoochee River population and stocked individuals from the lower Flint River population. Planning broodstock collections, translocations, and supplemental stocking efforts within genetic management units could better safeguard the existing genetic diversity within and among populations while also avoiding potential outbreeding depression. Similar principles apply to reintroduction efforts, wherein broodstock sourced from an encompassing population boundary or a nearby population may be better adapted to local selection pressures and less likely to induce outbreeding depression in dissimilar populations nearby.

Balancing the goals of genetic diversity conservation with sport fish management creates unique challenges for fisheries managers, who are tasked with sustaining quality fisheries while also maintaining robust populations, which implicitly includes genetic diversity. However, operating within population genetic management units could hinder stocking efforts due to a lack of suitable broodstock (e.g., low abundance or increased hybridization) or limited space and personnel dedicated to hatchery production. To address such challenges, management agencies and stakeholders are currently devising a rangewide management plan for Shoal Bass, complete with benchmark goals for conservation. The population structure revealed in this study will help to inform the management units used to implement that plan so that it best serves as an effective blueprint for management and restoration within a population genetic framework. A similar approach has been used recently with Florida Bass (Barthel et al. 2010; Porak et al. 2015) and Neosho Smallmouth Bass *M. dolomieu velox* (Taylor et al. 2016). Because supplemental stocking has the potential to affect Shoal Bass populations across jurisdictional boundaries, careful design of stocking programs will help to ensure that the needs of sport fish management are balanced with long-term conservation of genetic diversity within the species. Finally, additional research is needed to better understand how population structure is affected by both natural and anthropogenic factors to better illuminate management priorities.

ACKNOWLEDGMENTS

This research represents a collaborative effort undertaken by individuals affiliated with the U.S. Geological Survey (USGS), Oklahoma State University (OSU), the Florida Fish and Wildlife Conservation Commission's Fish and Wildlife Research Institute, GADNR, Auburn University, and the University of Georgia. Financial support for this publication was provided by the NPS Chattahoochee River National Recreation Area through the Great Plains Cooperative Ecosystems Studies Unit (Task Order P14AC00125). Additional funding was provided by the Otto S. Cox Graduate Fellowship for Genetics Research at OSU awarded to A.T.T. This study was performed under the auspices of the OSU Institutional Animal Care and Use Committee (Protocol Number AG-13-8). The Oklahoma Cooperative Fish and Wildlife Research Unit is supported by the Oklahoma Department of Wildlife Conservation, USGS, Wildlife Management Institute, and U.S. Fish and Wildlife Service. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government. There is no conflict of interest declared in this article.

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